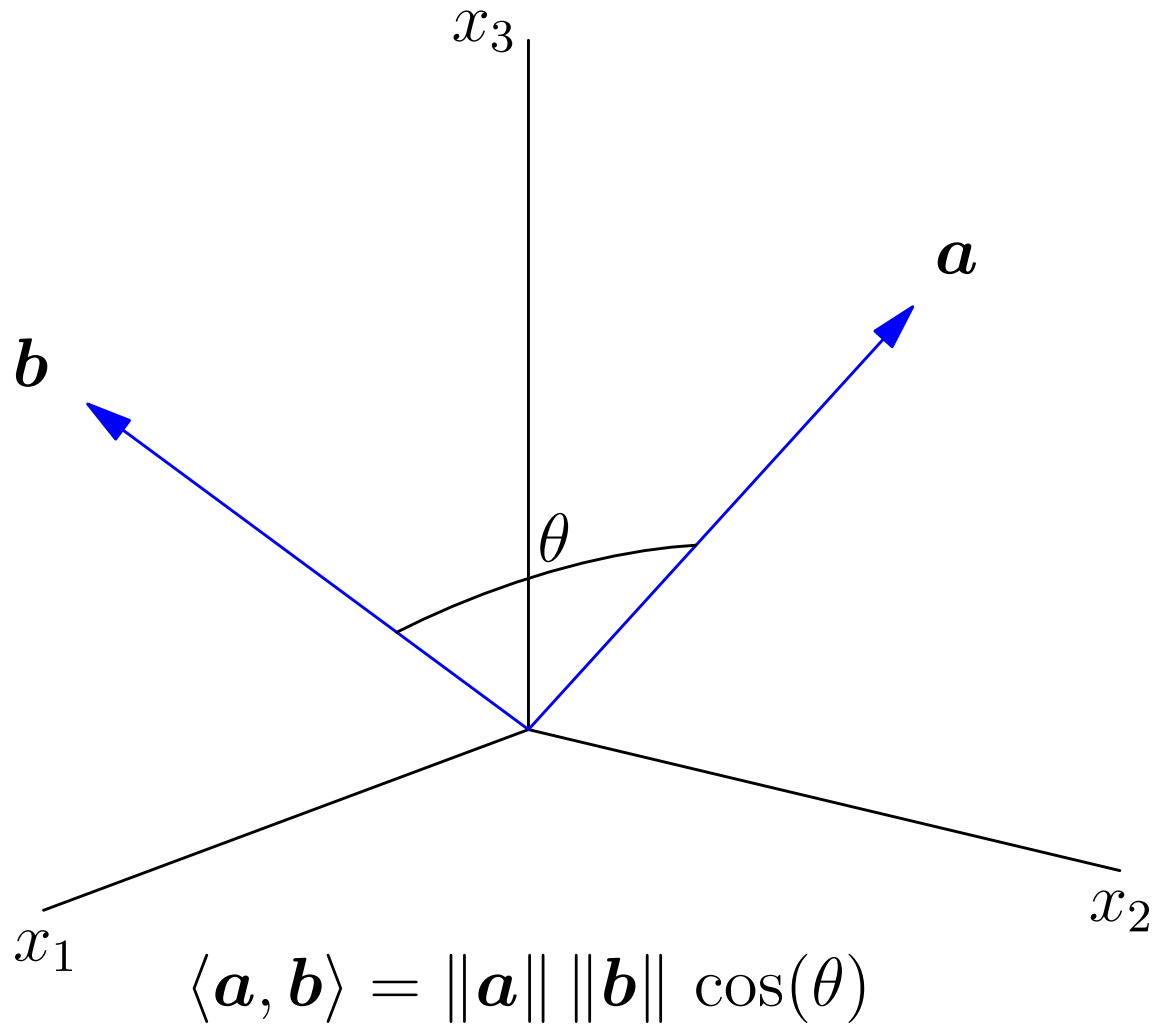


Mathematics of Machine Learning

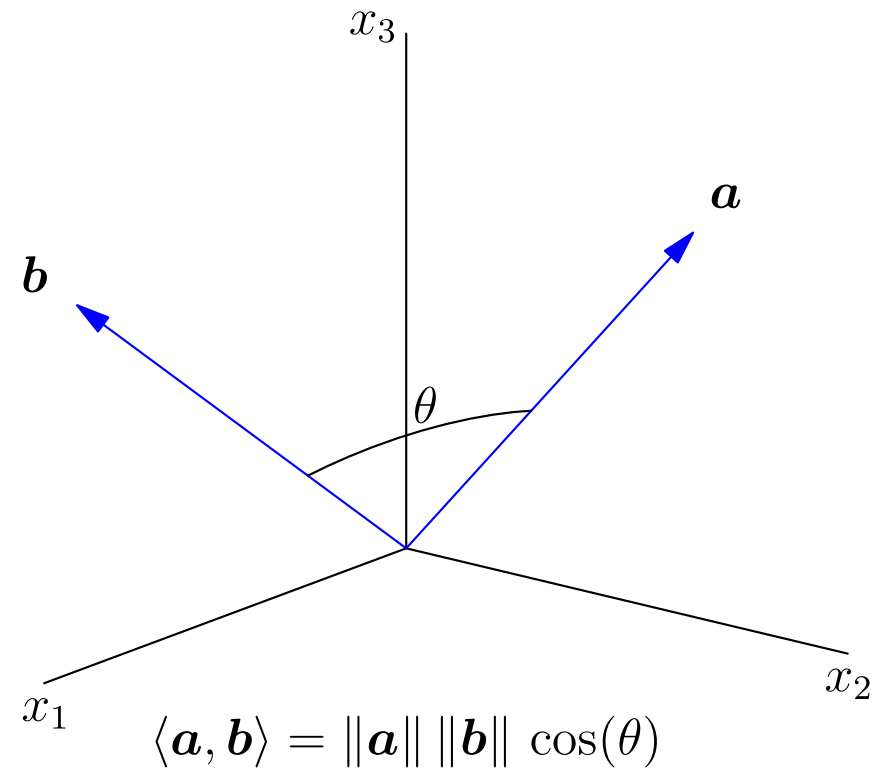
Inner Product Spaces



Inner products, operators

Outline

1. Inner Products
2. Operators



Recap

- We have looked at vector spaces (closed sets where we can add elements and multiply them by a scalar)
- Recall that vector spaces don't just apply to normal vectors ($\mathbb{R}^n$), but to matrices, functions, sequences, random variables, . . .
- Proper distances or metrics, $d(\mathbf{x}, \mathbf{y})$, allow us to construct ideas about geometry of the vector space
- Norms, $\|\mathbf{x}\|$, allow us to reason about the size of a vector
- Norms induce a distance, $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$

Inner Products

- We will often consider objects with an *inner product*, $\langle, \rangle$ ■
- For vectors in $\mathbb{R}^n$

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y} = \sum_{i=1}^n x_i y_i \blacksquare$$

- For functions

$$\langle f, g \rangle = \int_{x \in \mathcal{I}} f(x) g(x) dx \blacksquare$$

- For $m \times n$ matrices

$$\langle \mathbf{A}, \mathbf{B} \rangle = \text{Tr} \mathbf{A}^\top \mathbf{B} = \sum_{i=1}^m \sum_{j=1}^n A_{ij} B_{ij} \blacksquare$$

Axioms of Inner Products

- An inner product satisfies
 1. $\langle x, x \rangle \geq 0$ for all $x \in \mathcal{V}$ ■
 2. $\langle x, x \rangle = 0$ if and only if $x = \mathbf{0}$ ■
 3. $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle$ ■
 4. $\langle x, y + z \rangle = \langle x, y \rangle + \langle x, z \rangle$ ■
 5. $\langle x, y \rangle = \langle y, x \rangle$ ■
- We can show that $\|x\| = \sqrt{\langle x, x \rangle}$ satisfies the axioms of a norm, so that an inner-product space is a normed space■
- The norm associated with the inner-product for vectors in $\mathbb{R}^n$ (i.e. $\langle x, y \rangle = x^\top y$) is the Euclidean norm $\|x\| = \sqrt{x^\top x}$ ■

Cauchy-Schwarz Inequality

- One of the most important results of inner-product spaces, known as the **Cauchy-Schwarz inequality** is that

$$\langle \mathbf{x}, \mathbf{y} \rangle^2 \leq \langle \mathbf{x}, \mathbf{x} \rangle \langle \mathbf{y}, \mathbf{y} \rangle = \|\mathbf{x}\|^2 \|\mathbf{y}\|^2$$

- Or

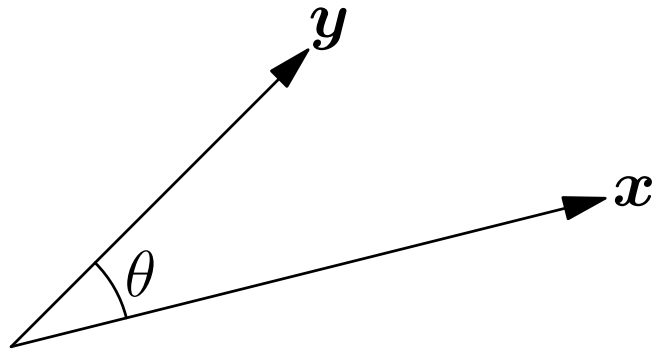
$$|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \|\mathbf{y}\|$$

- This is a very general result so for example

$$\left| \int f(x)g(x)dx \right| \leq \sqrt{\left(\int f^2(x)dx \right) \left(\int g^2(x)dx \right)}$$

Angles Between Vectors

- A natural interpretation of the inner product is in providing a measure of the angle between vectors■



$$\langle x, y \rangle = x^T y = \|x\| \|y\| \cos(\theta)$$

- Vectors are orthogonal if $\langle x, y \rangle = 0$ ■
- We can extend this idea to functions

$$\langle f(x), g(x) \rangle = \int_{x \in \mathcal{I}} f(x) g(x) dx = \|f(x)\| \|g(x)\| \cos(\theta) \blacksquare$$

- Note that $\sin(x)$ and $\cos(x)$ are orthogonal in the interval $[0, 2\pi]$ ■

Basis Functions

- Any set of vectors $\{\mathbf{b}_i | i = 1, \dots\}$ that span the space can be used as a basis or coordinate system■
- The simplest and most useful case is when the vectors are orthogonal and normalised (i.e. $\|\mathbf{b}_i\| = 1$)■

- In $\mathbb{R}^3$ we could use $\mathbf{b}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\mathbf{b}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, $\mathbf{b}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ ■

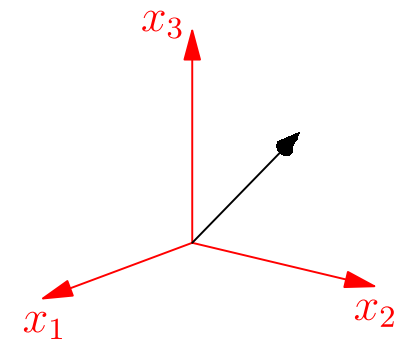
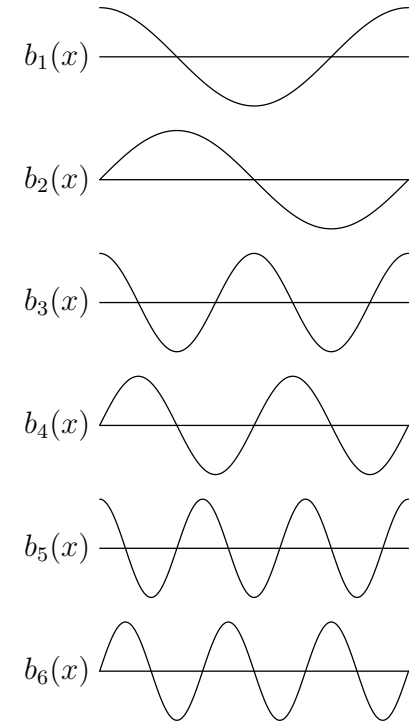
- This is not unique as we can rotate our basis vectors■

- For an orthonormal basis we can write any vector as

$$\hat{\mathbf{x}} = \begin{pmatrix} \mathbf{x}^\top \mathbf{b}_1 \\ \mathbf{x}^\top \mathbf{b}_2 \\ \mathbf{x}^\top \mathbf{b}_3 \end{pmatrix} \quad \blacksquare$$

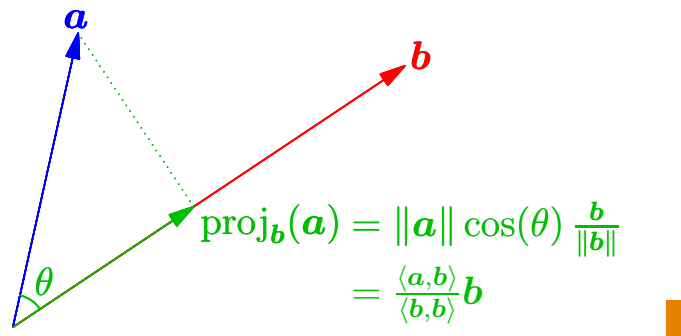
Orthogonal Functions

- For functions we can use any ortho-normal set of functions as a basis■
- The most familiar are the Fourier functions $\sin(n\theta)$ and $\cos(n\theta)$ ■
- Any function in $C(0,2\pi)$ can be represented by a point $\mathbf{f} = \begin{pmatrix} \langle f(x), b_0(x) \rangle \\ \langle f(x), b_1(x) \rangle \\ \vdots \end{pmatrix}$ ■
- There might be an infinite number of components■
- This is analogous to points in $\mathbb{R}^n$ (for large n)■



Projection Operator

- Given a vector a we can project it onto a vector b

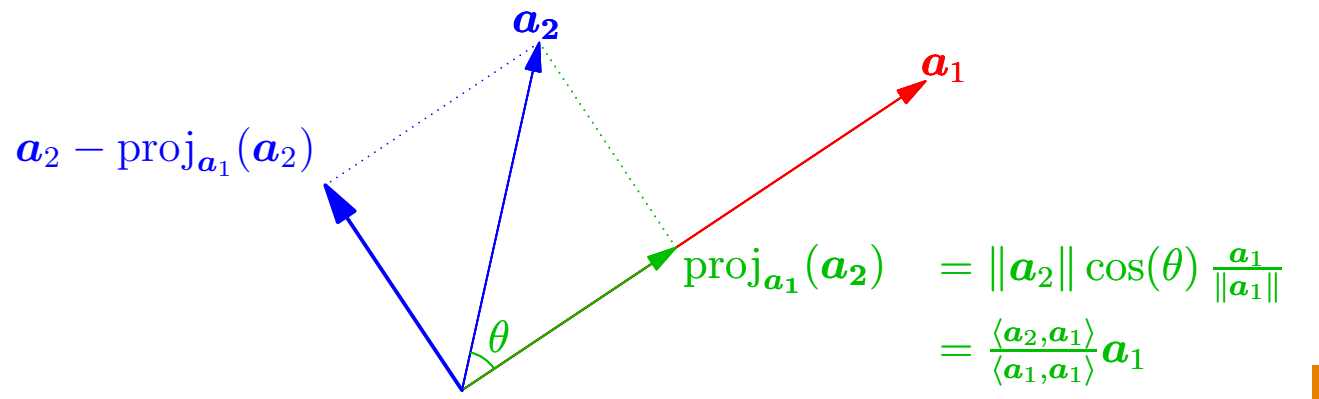


- Using $\langle a, b \rangle = \|a\| \|b\| \cos(\theta)$ and $\langle b, b \rangle = \|b\|^2$
- A defining feature of a projection operator is

$$\begin{aligned}\text{proj}_b(\text{proj}_b(a)) &= \text{proj}_b\left(\frac{\langle a, b \rangle}{\langle b, b \rangle} b\right) \\ &= \left\langle \frac{\langle a, b \rangle}{\langle b, b \rangle} b, b \right\rangle \frac{b}{\langle b, b \rangle} = \frac{\langle a, b \rangle}{\langle b, b \rangle} b = \text{proj}_b(a)\end{aligned}$$

Constructing an Orthonormal Basis

- Given a set of linearly independent vectors $(\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n)$ we can construct an orthonormal basis



- We can iterate this

$$\mathbf{u}_1 = \mathbf{a}_1$$

$$\mathbf{e}_1 = \frac{\mathbf{u}_1}{\|\mathbf{u}_1\|}$$

$$\mathbf{u}_2 = \mathbf{a}_2 - \text{proj}_{\mathbf{u}_1}(\mathbf{a}_2)$$

$$\mathbf{e}_2 = \frac{\mathbf{u}_2}{\|\mathbf{u}_2\|}$$

$$\mathbf{u}_3 = \mathbf{a}_3 - \text{proj}_{\mathbf{u}_1}(\mathbf{a}_3) - \text{proj}_{\mathbf{u}_2}(\mathbf{a}_3)$$

$$\mathbf{e}_3 = \frac{\mathbf{u}_3}{\|\mathbf{u}_3\|}$$

Gram-Schmidt Orthogonalisation

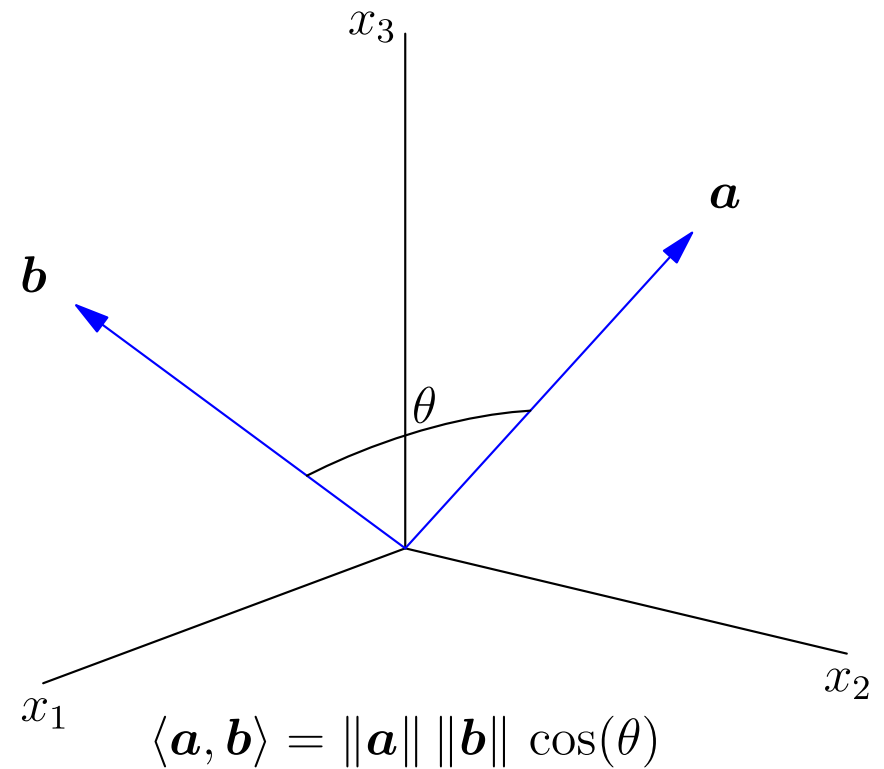
- The algorithm in the last slide is known as **Gram-Schmidt**
- We can perform it on any inner-product space
- For vectors in $\mathbb{R}^n$ it is prone to accumulate rounding errors and is not the method of choice
- Numerical packages use “Householder transformations” to perform orthogonalisation
- Nevertheless, it is nice to know that we can form an orthonormal basis

Algebraic Structure

- We have gone to these lengths as we want to show that many properties of vectors are shared by other objects (matrices, functions, etc.)■
- The notions of distance (geometry), norms (size of vectors) and inner products (angles between vectors) provides a very rich set of concepts■
- Vectors form the backbone of objects we will use repeatedly in machine learning■
- The next piece of the jigsaw is to understand how we can transform these objects■

Outline

1. Inner Products
2. **Operators**



Operators

- In machine learning we are interested in transforming our input vectors into some output predictions■
- To accomplish this we will apply some mapping or operators on the vector $\mathcal{T} : \mathcal{V} \rightarrow \mathcal{V}'$ ■
- This says that $\mathcal{T}$ maps some object $x \in \mathcal{V}$ to an object $y = \mathcal{T}[x]$ in a new vector space $\mathcal{V}'$ ■
- This new vector space may or may not be the same as the original vector space■
- Our objects may be any object in a vector space such as a function■

Linear Operators

- Operators are in general very complicated, but a particularly nice set of operators are linear operators■
- $\mathcal{T}$ is a linear operator if
 1. $\mathcal{T}[a\mathbf{x}] = a\mathcal{T}[\mathbf{x}]$
 2. $\mathcal{T}[\mathbf{x} + \mathbf{y}] = \mathcal{T}[\mathbf{x}] + \mathcal{T}[\mathbf{y}]$ ■
- For normal vectors ($\mathbf{x} \in \mathbb{R}^n$) the most general linear operation is

$$\mathcal{T}[\mathbf{x}] = \mathbf{M}\mathbf{x}$$

where $\mathbf{M}$ is a matrix■

Matrix multiplication

- For an $\ell \times m$ matrix $\mathbf{A}$ and an $m \times n$ matrix $\mathbf{B}$ we can compute the $(\ell \times n)$ product, $\mathbf{C} = \mathbf{AB}$, such that

$$C_{ij} = \sum_{k=1}^m A_{ik} B_{kj} \quad \left(\begin{array}{|c|} \hline \text{---} \\ \hline \text{---} \\ \hline \text{---} \\ \hline \end{array} \right) \left(\begin{array}{|c|c|c|} \hline | & | & | \\ \hline | & | & | \\ \hline | & | & | \\ \hline \end{array} \right) = \left(\begin{array}{|c|c|c|} \hline \blacksquare & \blacksquare & \blacksquare \\ \hline \blacksquare & \blacksquare & \blacksquare \\ \hline \blacksquare & \blacksquare & \blacksquare \\ \hline \end{array} \right)$$

- Treating the vector $\mathbf{x}$ as an $n \times 1$ matrix then

$$\mathbf{y} = \mathbf{Ax} \quad \Rightarrow \quad y_i = \sum_j A_{ij} x_j \quad \left(\begin{array}{|c|} \hline \text{---} \\ \hline \text{---} \\ \hline \text{---} \\ \hline \end{array} \right) \left(\begin{array}{|c|} \hline | \\ \hline | \\ \hline | \\ \hline \end{array} \right) = \left(\begin{array}{|c|} \hline \blacksquare \\ \hline \blacksquare \\ \hline \blacksquare \\ \hline \end{array} \right)$$

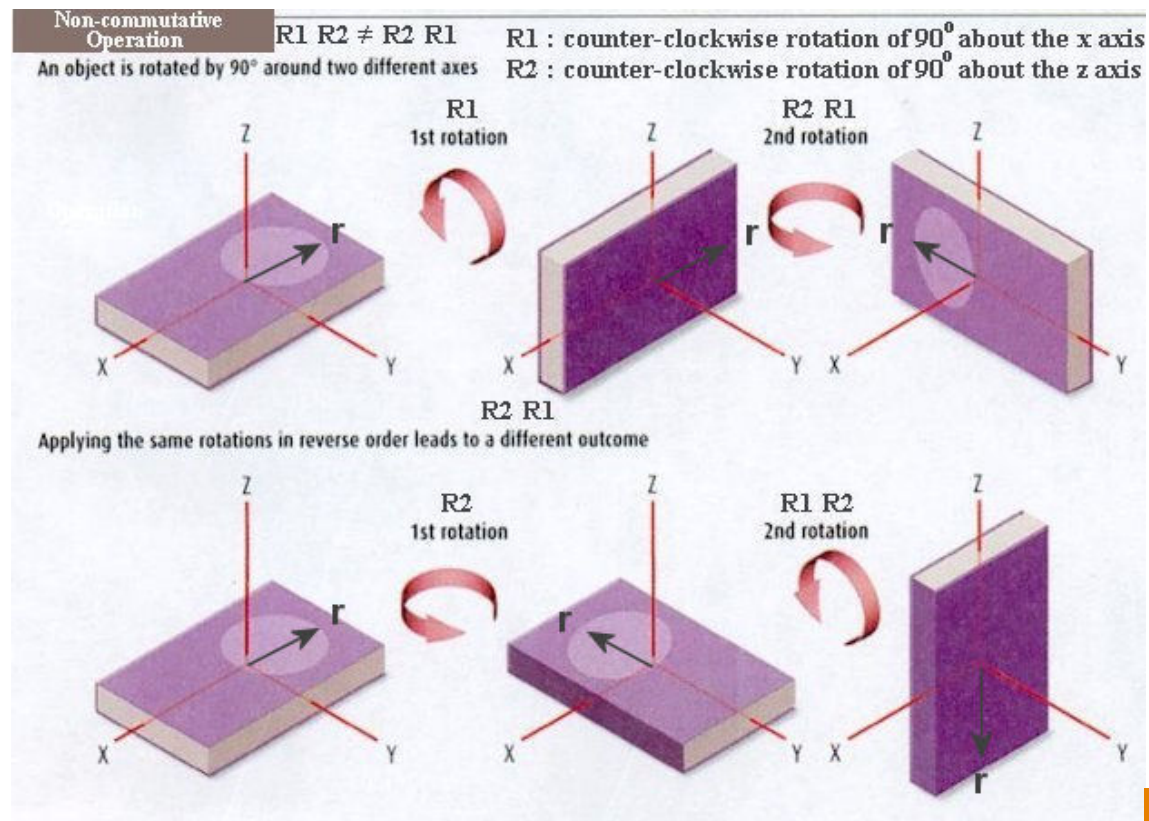
- Using the same matrix notation we can define the inner product as

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{y} = \sum_{i=1}^n x_i y_i \quad \left(\text{---} \right) \left(\begin{array}{|c|} \hline | \\ \hline | \\ \hline | \\ \hline \end{array} \right) = \left(\blacksquare \right)$$

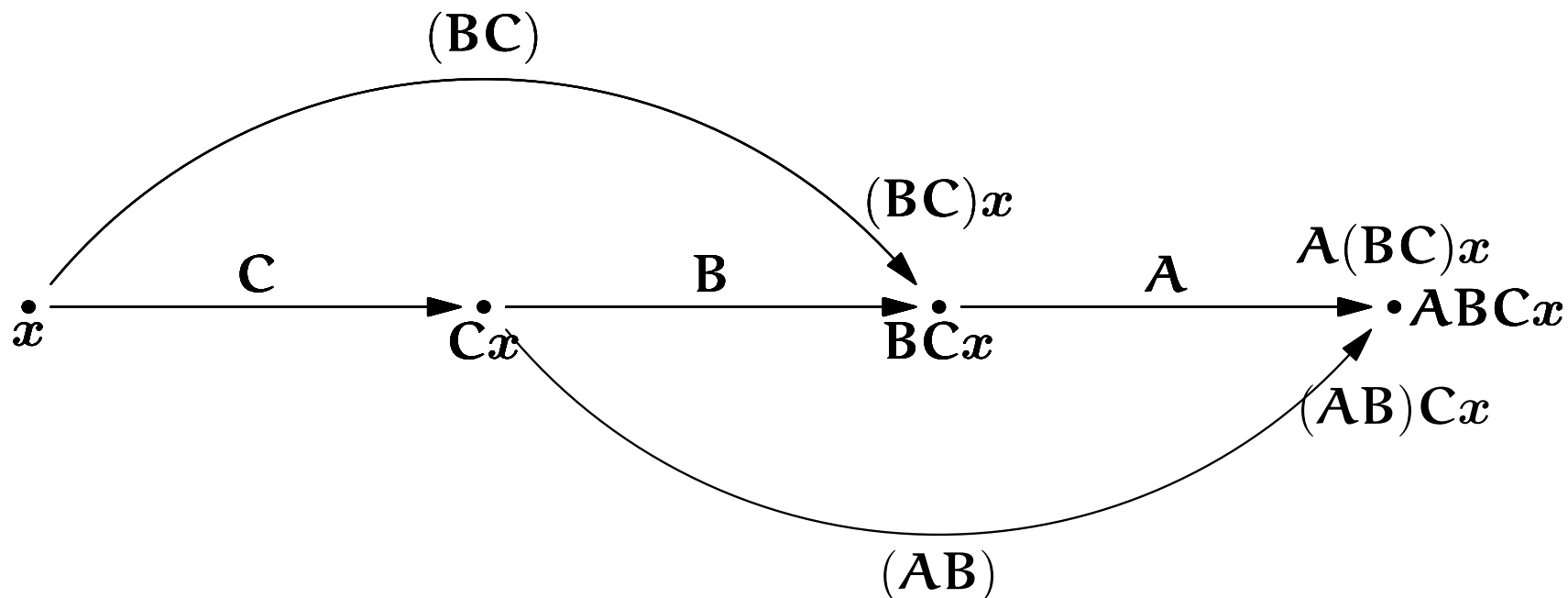
Non-commutativity

- In general $AB \neq BA$

$$\begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{pmatrix}$$



Associativity of Mappings



- For all x we have $A(BC)x = (AB)Cx$
- This implies $A(BC) = (AB)C$

Kernels

- The equivalent of a matrix for functions (i.e. a linear operator) is known as a kernel $K(x, y)$

$$g(x) = \mathcal{T}[f] = \int_{y \in \mathcal{I}} K(x, y) f(y) dy$$

- Our domain does not need to be one dimensional, e.g.

$$g(\boldsymbol{x}) = \mathcal{T}[f] = \int_{\boldsymbol{y} \in \mathcal{I}} K(\boldsymbol{x}, \boldsymbol{y}) f(\boldsymbol{y}) d\boldsymbol{y}$$

- We shall soon see examples of high-dimensional kernels

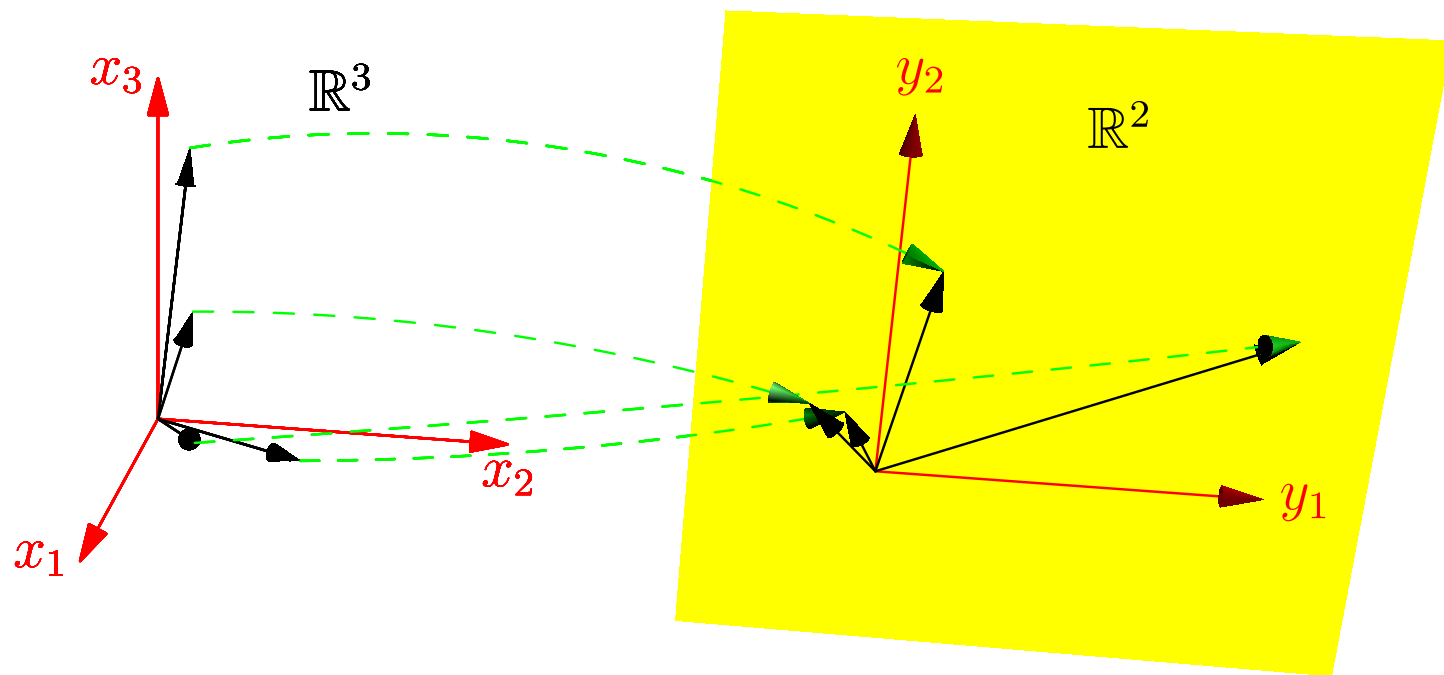
Kernels in Machine Learning

- Kernels are used heavily in machine learning■
- In kernel methods such as SVM, SVR, Kernel-PCA■
- They are also used in Gaussian Processes■
- In all these cases we consider symmetric, positive semi-definite kernels■
- Sometimes they can be interpreted as covariance between random functions

$$K(\mathbf{x}, \mathbf{y}) = \mathbb{E}_{f \sim \mathcal{P}} \left[(f(\mathbf{x}) - \mu(\mathbf{x})) (f(\mathbf{y}) - \mu(\mathbf{y})) \right] \blacksquare$$

General Linear Mappings

- In general a linear operator will map vectors between different vector spaces
- E.g. $\mathbb{R}^3 \rightarrow \mathbb{R}^2$



Square Matrices

- We will spend a lot of time on operators that map from a vector space onto itself $\mathcal{T} : \mathcal{V} \rightarrow \mathcal{V}$
- For vectors in $\mathbb{R}^n$ such linear operators are represented by square matrices
- When there is a one-to-one mapping then we have a unique inverse
- We will study such mappings in detail in the next lecture

Summary

- Mathematics is the language of machine learning and you have to get used to it■
- Mathematics is like programming, if you don't understand the syntax and you can't write it down then it's meaningless■
- We've taken a high level view of inner product spaces and operators, this will pay us back later as we look at kernel methods■